

$$\Phi_R(s) = \frac{KG_1(s)G_2(s)}{1+KG_1(s)G_2(s)}$$

tracking path: command to output

$$\Phi_N(s) = \frac{G_2(s)}{1+KG_1(s)G_2(s)}$$

disturbance path: wave to output

$$y(t) = y_R(t) + y_N(t)$$

$$C_N(\infty) \approx \frac{0.625}{K} \text{ for step disturbance}$$